Sets

Propositional logic is about reasoning, and inferring new facts from known facts.

But we also need to model different sorts of data in order to reason about it.

The most basic data structure is a ***set***, which is simply a collection of objects.

These objects typically share a property. For example:

- The set of students enrolled in CS-170.
- The set of (named) beaches on the Gower Peninsula.
- The set of programs in a particular computer library.

The objects of a set are referred to as its ***elements***, or ***members***.

The ***cardinality*** (***size***) of a set A, denoted by $|A|$, is the number of objects in A.

Set Notation

Sets are depicted by listing their elements within curly braces; for example:

- $\{ 3, 7, 13 \}$
- $\{ \text{red}, \text{blue}, \text{yellow} \}$
- $\{ \text{Joel}, \text{Felix}, \text{Oskar}, \text{Amanda} \}$
- $\{ 1, 3, 5, \ldots, 99 \}$
- $\{ \text{a}, \text{b}, \text{c}, \ldots, \text{z} \}$
- $\{ 2, 3, 5, 7, 11, 13, 17, \ldots \}$

The sets on the right, with the ellipses, may be ambiguous. To avoid this, we use explicit ***set building notation***:

$$\{ x : x \text{ has property } P \}$$

This denotes the collection of exactly those objects x which satisfy property P. For example:

- $\{ b : b \text{ is a beach on Gower Peninsula} \}$
- $\{ p : p \text{ has climbed Mount Kailash} \}$
- $\{ n : n \text{ is a prime number} \}$
- $\{ A : A \text{ is a set of people who share a common grandmother} \}$

Example Sets

The following are common sets used in mathematics:

$$\emptyset = \{\,\} \qquad \text{(the empty set)}$$

$$\mathbb{B} = \{0, 1\} \qquad \text{(binary digits, or bits)}$$

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\} \qquad \text{(natural numbers)}$$

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\} \qquad \text{(integers)}$$

$$\mathbb{Q} = \left\{\frac{m}{n} : m, n \in \mathbb{Z},\ n \neq 0\right\} \qquad \text{(rational numbers)}$$

$$\mathbb{R} = \{x : x \text{ is a real number}\} \qquad \text{(real numbers)}$$

Note: $\emptyset$ and $\{\emptyset\}$ are very different sets!

$\emptyset = \{\,\}$ contains zero elements – it is empty.

$\{\emptyset\}$ contains one element – it is *not* empty.

Example Sets

Exercise: Write out the following sets explicitly.

1. $\{ x : x \text{ is an odd integer with } 0 < x < 8 \}$

2. $\{ p : p \text{ is an even prime number} \}$

3. $\{ x : x \text{ is a day of the week not containing the letter n} \}$

4. $\{ d : d \text{ is a nephew of Donald Duck} \}$

Membership of a Set

The symbol $\in$ is used to represent *membership* of a set:

$x \in A$ says "x is an *element* (or a *member*) of A".

$x \notin A$ is the same as $\neg (x \in A)$.

$x \notin A$ says "x is *not* an element of A".

Examples:

$$7 \in \{ 3, 7, 14 \}$$

$$\text{Felix} \in \{ \text{Joel, Felix, Oskar} \}$$

$$8 \notin \{ 3, 7, 14 \}$$

$$\text{Amanda} \notin \{ \text{Joel, Felix, Oskar} \}$$

Set Equality

Two sets are ***equal*** if, and only if, they have the same elements.

That is to say, $A = B$ if for any x:

$$x \in A \quad \text{if, and only if,} \quad x \in B.$$

Note: When writing out a set, it doesn't matter in what order you list the elements, nor how many times you list the elements.

Examples:

$$\{3, 7, 14\} = \{7, 14, 3, 7, 3\}$$

$$\{\text{Joel, Felix, Oskar}\} \neq \{\text{Joel, Felix, Oskar, Amanda}\}$$

Set Membership and Equality

Exercise: Which of the following propositions are true?

1. $2 \in \{1, 2, 3\}$
2. $\{2\} \in \{1, 2, 3\}$
3. $\{2\} \in \{\{1\}, \{2\}, \{3\}\}$
4. $\emptyset \in \{\}$
5. $\emptyset \in \{\emptyset\}$

Exercise: Which of the following sets are equal?

1. $A = \{1, \{1, 2\}\}$
2. $B = \{1, \{2\}\}$
3. $C = \{1, \{1\}\}$
4. $D = \{\{1, 1\}, 1\}$
5. $E = \{\{2, 1\}, 1\}$

Subsets

Set A is a ***subset*** of set B, if, and only if, each element of A is an element of B; in such case we write $A \subseteq B$.

We also say: B is a ***superset*** of A, written $B \supseteq A$; A is *included*, or *contained*, in B; and B *includes*, or *contains*, A.

$A \not\subseteq B$ denotes that A is *not* a subset of B, which means there is an element in A which is not in B. This is the same as $\neg(A \subseteq B)$.

$A \subset B$ denotes that A is a ***proper subset*** of B; that is $A \subseteq B$ but $A \neq B$.

Example: $\emptyset \subset \mathbb{B} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

Example: $\{2\} \subseteq \{1,2,3\}$

$\{1,2,3\} \not\subseteq \{\{1\},\{2\},\{3\}\}$

$\{\{1,2\}\} \not\subseteq \{\{1,2,3\}\}$

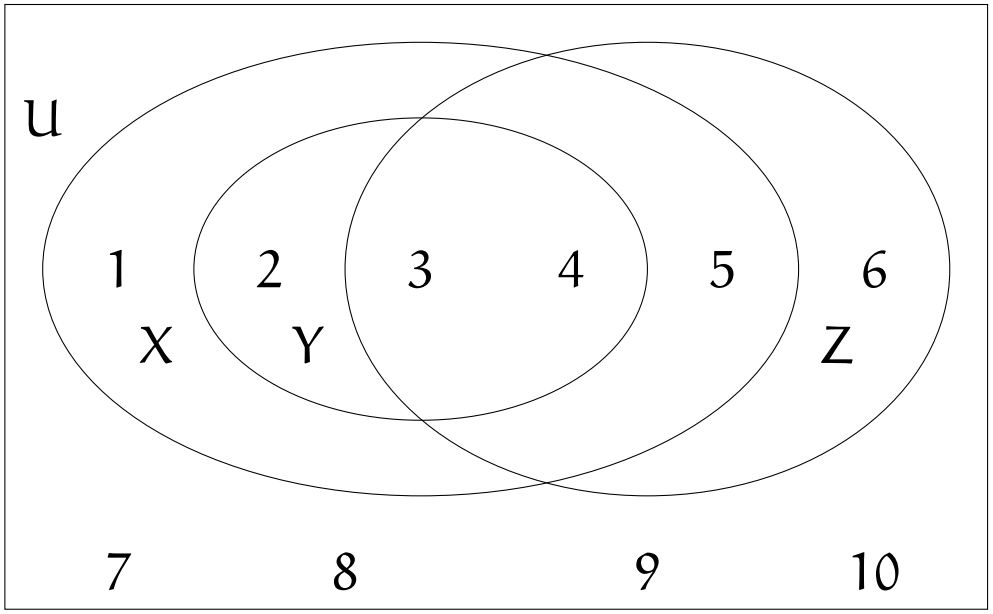
Venn Diagrams

Venn diagrams are a graphical way of depicting sets.

Consider the sets $X = \{1, 2, 3, 4, 5\}$, $Y = \{2, 3, 4\}$, and $Z = \{3, 4, 5, 6\}$ of objects taken from some universal set U, referred to as the ***universe of discourse***; in this case:

$$U = \{\, 1,\, 2,\, \ldots,\, 9,\, 10 \,\}$$

These sets can be depicted as follows.



We can explore the relationships between these three sets.

- $Y \subseteq X$; moreover
- $Y \subset X$ since $1 \in X$ but $1 \notin Y$
- $X \not\subseteq Z$ and $Z \not\subseteq X$
- $Y \not\subseteq Z$ and $Z \not\subseteq Y$

Venn Diagrams

Using Venn diagrams we easily see that, for any sets A, B, C over a universe of discourse U:

- $\emptyset \subseteq A$ and $A \subseteq U$.

That is: $\emptyset$ is the *smallest* set with respect to inclusion; and U is the *largest* set with respect to inclusion.

Using Venn diagrams we also easily note the following properties of the subset relation $\subseteq$:

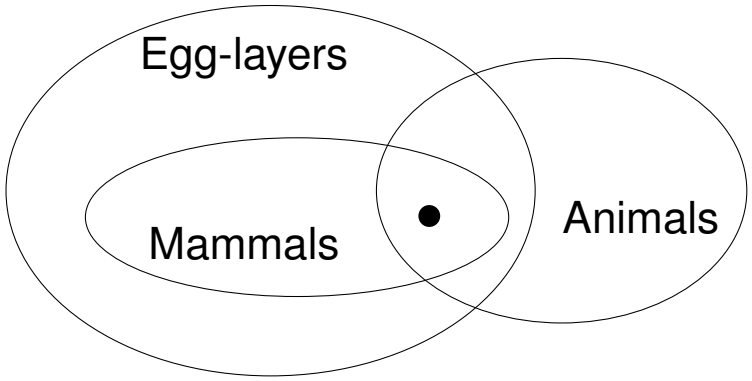
- It is ***reflexive***, meaning that: $A \subseteq A$.
- It is ***antisymmetric***, meaning that:

  if $A \subseteq B$ and $B \subseteq A$ then $A = B$.
- It is ***transitive***, meaning that:

  if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

Modelling with Sets

Here is a Venn diagram illustrating the validity of the following deduction:

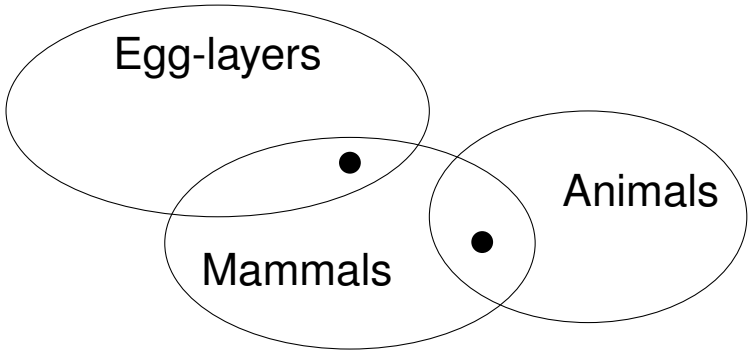


Some animals are mammals.

All mammals lay eggs.

Therefore, some animals lay eggs.

Here is a Venn diagram illustrating the invalidity of the following deduction:



Some animals are mammals.

Some mammals lay eggs.

Therefore, some animals lay eggs.

Modelling with Sets

Exercise: A Lewis Carroll puzzle assumes the following:

- All babies are illogical.
- Nobody is despised who can manage a crocodile
- Illogical persons are despised.

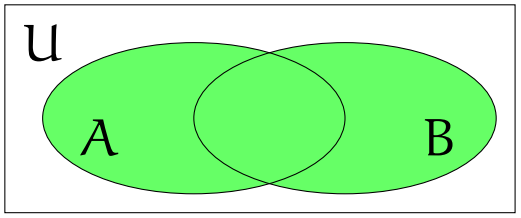
We are to deduce that no baby can manage a crocodile, using an appropriate Venn diagram.

Exercise: Use an appropriate Venn diagram to determine whether or not the following argument is valid.

- All oceans are full of water.
- No ponds are oceans.
- Therefore, no ponds are full of water.

Set Union

The **union** of two sets is the set consisting of all of the elements of the two sets taken together. Pictorially:



$A \cup B$

That is: $A \cup B = \{ x : x \in A \text{ or } x \in B \}$

Thus: $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$

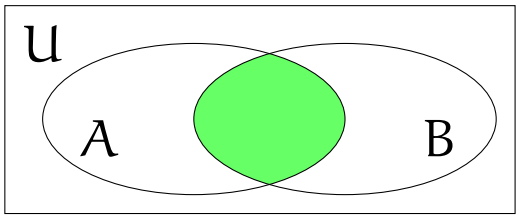
Example:

If $A = \{1, 2, 3, 4, 5\}$
$B = \{2, 4, 6, 8\}$
$C = \{3, 6, 9\}$

then $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$
$A \cup C = \{1, 2, 3, 4, 5, 6, 9\}$
$B \cup C = \{2, 3, 4, 6, 8, 9\}$
$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 8, 9\}$

Set Intersection

The **intersection** of two sets is the set consisting of all of the elements that are in *both* of the two sets. Pictorially:



$A \cap B$

That is: $A \cap B = \{ x : x \in A \text{ and } x \in B \}$

Thus: $x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$

Example:

If $A = \{1, 2, 3, 4, 5\}$ then $A \cap B = \{2, 4\}$

$B = \{2, 4, 6, 8\}$ $A \cap C = \{3\}$

$C = \{3, 6, 9\}$ $B \cap C = \{6\}$

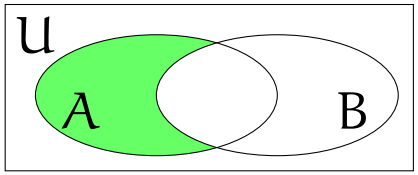
$A \cap B \cap C = \emptyset$

Inclusion-Exclusion Principle: For finite sets A and B:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Set Difference

The **difference** of two sets is the set consisting of all of the elements of the first that are *not* in the second. Pictorially:



$A \setminus B$

That is: $A \setminus B = \{ x : x \in A \text{ and } x \notin B \}$

Thus: $x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B$

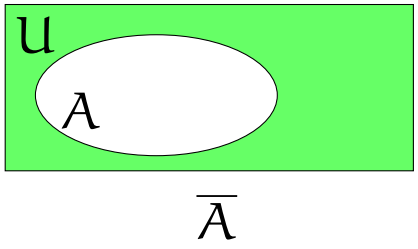
Example:

If $A = \{1, 2, 3, 4, 5\}$
$B = \{2, 4, 6, 8\}$
$C = \{3, 6, 9\}$

then $A \setminus B = \{1, 3, 5\}$
$B \setminus A = \{6, 8\}$
$A \setminus C = \{1, 2, 4, 5\}$
$C \setminus A = \{6, 9\}$
$B \setminus C = \{2, 4, 8\}$
$C \setminus B = \{3, 9\}$

Set Complement

The **complement** of a set is the set consisting of all of the elements that are *not* in the set. Pictorially:



That is: $\overline{A} = \{ x : x \notin A \}$

Thus: $x \in \overline{A} \iff x \notin A$

Note: the complement operation is taken with respect to an underlying universe of discourse U. That is, $\overline{A} = U \setminus A$.

Example: Assume $U = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9 \}$.

If $A = \{ 1, 2, 3, 4, 5 \}$ then $\overline{A} = \{ 6, 7, 8, 9 \}$

$B = \{ 2, 4, 6, 8 \}$ $\overline{B} = \{ 1, 3, 5, 7, 9 \}$

$C = \{ 3, 6, 9 \}$ $\overline{C} = \{ 1, 2, 4, 5, 7, 8 \}$

Set Operations

Exercise: Consider the following sets A, B and C over the the univere of discourse $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

$$A = \{1, 3, 5, 7, 9\}$$
$$B = \{3, 4, 5\}$$
$$C = \{5, 6, 7, 8, 9\}$$

Draw a Venn diagram depicting these sets, and compute the following sets:

1. $A \cap B$
2. $(A \cap B) \cup C$
3. $A \cap (B \cup C)$
4. $(A \cup B) \setminus C$
5. $\overline{(A \cup B)} \cap C$

Powerset

The ***powerset*** $\mathcal{P}(A)$ of a set A is the set consisting of all subsets of A.

That is: $\mathcal{P}(A) = \{ X : X \subseteq A \}$ In particular:

Thus: $x \in \mathcal{P}(A) \Leftrightarrow x \subseteq A$ $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$

Examples:

1. $\mathcal{P}(\{0, 1\}) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$
2. $\mathcal{P}(\{\text{Joel, Felix, Oskar, Amanda}\})$
   $= \{\emptyset, \{\text{Joel}\}, \{\text{Felix}\}, \{\text{Oskar}\}, \{\text{Amanda}\},$
   $\{\text{Joel, Felix}\}, \{\text{Joel, Oskar}\}, \{\text{Joel, Amanda}\},$
   $\{\text{Felix, Oskar}\}, \{\text{Felix, Amanda}\}, \{\text{Oskar, Amanda}\},$
   $\{\text{Joel, Felix, Oskar}\}, \{\text{Joel, Felix, Amanda}\},$
   $\{\text{Joel, Oskar, Amanda}\}, \{\text{Felix, Oskar, Amanda}\},$
   $\{\text{Joel, Felix, Oskar, Amanda}\}\}$
3. $\mathcal{P}(\emptyset) = \{\emptyset\}$ and $\mathcal{P}(\{a\}) = \{\emptyset, \{a\}\}$

Note: If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$

Sets and Properties

We typically use set-building notation $\{ x : x \text{ has property } P \}$ to define a subset of a given set A, in which case we write:

$$B = \{ x \in A : x \text{ has property } P \}$$

instead of $B = \{ x : x \in A \text{ and } x \text{ has } P \}$

Example: Instead of defining set difference as

$$A \setminus B = \{ x : x \in A \text{ and } x \notin B \}$$

we can write

$$A \setminus B = \{ x \in A : x \notin B \}$$

Russell's Paradox

Set-building notation is useful, but must be used with care.

Say a set A is *normal* if it is not an element of itself: $A \notin A$.

Consider the set R of all normal sets: $R = \{ A : A \notin A \}$.

Is R normal? That is, is the statement $R \in R$ true or false?

- If $R \notin R$, then R has the property which defines R,
  which would mean that $R \in R$.
- If $R \in R$, then R does *not* have the property which defines R,
  which would mean that $R \notin R$.

This anomaly is known as *Russell's Paradox*, after Bertrand Russell.

If we use set-building notation only in restricted form:

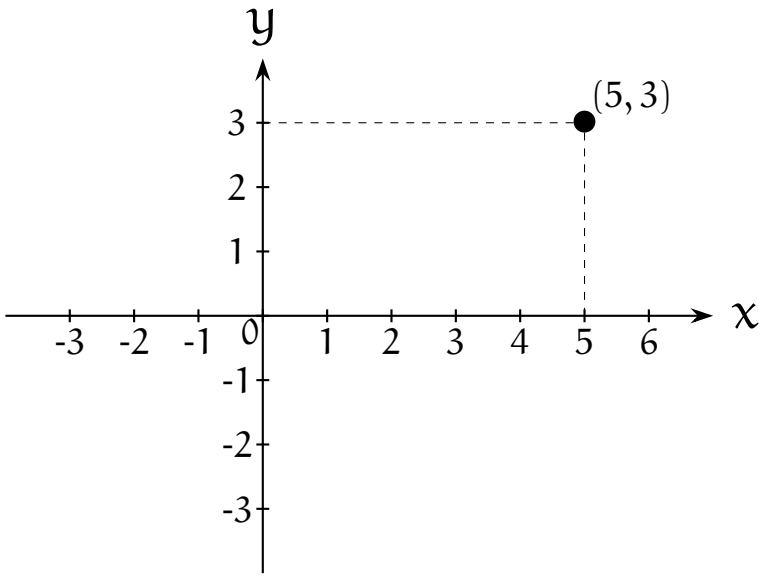
$$R = \{ x \in A : x \text{ has property } P \}$$

then we will never encounter such anomalies.

Ordered Pairs, Triples, n-tuples

An ***(ordered) pair*** is a comma-separated pair of objects (a, b) written within parantheses. The objects a and b are referred to as *coordinates*.

Example: Points in the xy-plane are denoted by ordered pairs; e.g., the pair $(5, 3)$ refers to the point corresponding to 5 on the horizontal (x) axis and 3 on the vertical (y) axis.



Two pairs are equal if, and only if, their respective coordinates are equal:

$$(a, b) = (c, d) \quad \Leftrightarrow \quad a = c \ \wedge \ b = d$$

We can also have ***triples***, and more generally ***n-tuples*** consisting of an arbitrary number n of coordinates.

Cartesian Products

Given sets A and B, the set of all pairs (a, b) with $a \in A$ and $b \in B$ is denoted:

$$A \times B \;=\; \big\{\, (a, b) :\; a \in A \text{ and } b \in B \,\big\}$$

Thus: $(a, b) \in A \times B \;\Leftrightarrow\; a \in A \;\wedge\; b \in B$

Example:

The Cartesian product

$$\{1, 2, \ldots, m\} \;\times\; \{1, 2, \ldots, n\}$$

can model a finite grid, such as the points of an LCD screen or the squares of a chess board.

$$\begin{array}{cccc} (1,1) & (1,2) & \ldots & (1,n) \\ (2,1) & (2,2) & \ldots & (2,n) \\ \vdots & \vdots & \ddots & \vdots \\ (m,1) & (m,2) & \ldots & (m,n) \end{array}$$

Note: The number of elements in a Cartesian product is the product of the number of elements of the individual sets.

$$|A \times B| \;=\; |A| \times |B|.$$

In particular, $A \times \emptyset \;=\; \emptyset \times A \;=\; \emptyset$.

Cartesian Products

We can form the Cartesian product of any number of sets:

$$A_1 \times A_2 \times \cdots \times A_n = \{ (a_1, a_2, \ldots, a_n) : a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n \}$$

We write A^n for $A \times A \times \ldots \times A$.

Example:

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ represents the two-dimensional plane.

$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ represents three-dimensional space.

$\mathbb{Z}^n = \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ represents the set of n-tuples of integers.

Cartesian Products

Example: Let S represent all students, C all courses, and G all possible grades.

Then $S \times C \times G$ represents all triples (s, c, g) where $s \in S$ is a student, $c \in C$ is a course, and $g \in G$ is a grade.

A University student database would be a subset of $S \times C \times G$ recording the grades for all students registered in each course.

Example: An address book would be represented by a subset of

$$\mathit{Names} \times \mathit{Addresses} \times \mathit{Phones}$$

where *Names*, *Addresses* and *Phones* are the sets of all possible Names, Addresses and Phone Numbers.